

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):		
$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)	
$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)	
$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)	
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)		
Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.		
ê by polarization plug into templates above		
Pol.	ê	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{ } = (1 + \Gamma_{ }) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{ } ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_2}{\eta_1}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{ } ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R+T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{ } = 1 - R_{ }$
Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; <i>nonmag.</i> ⇒ $\eta_2 / \eta_1 = n_1 / n_2$.			
Intrinsic impedance η_i plug into table above			
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$			
Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says “ferromagnetic”/iron/ferrite/etc.			
Low-loss correction $\eta_c \sigma / (\omega \epsilon) \ll 1$			
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega \epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$			
For Γ/τ just use $\eta_0 / \sqrt{\epsilon_r}$. The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.			

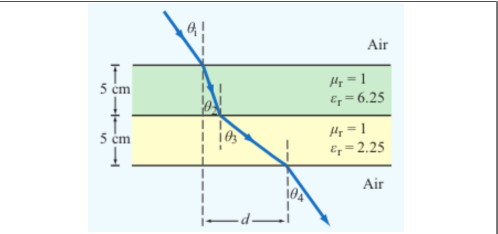
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	
$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$	
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	
$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$	
Air-slabs-air shortcut <i>outer media identical</i>	
$\theta_{\text{exit}} = \theta_{\text{incident}}$	
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$		
Type	$\sigma / \omega \epsilon$	η_c, α, β
Lossless (σ = 0)	0	$\eta = \eta_0 / \sqrt{\epsilon_r}$ exact $\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \epsilon'}{2\omega \epsilon_r}\right)$ $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	$10^{-2} - 10^2$	exact: $(\eta / \sqrt{X}) e^{j\theta_\eta}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j) \sqrt{\pi f \mu / \sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r / \epsilon_r} \eta_0$

For Γ/τ: drop j correction; use $\eta_0 / \sqrt{\epsilon_r}$. Magnetic: keep μ_r.

Low-loss quick params $\sigma / (\omega \epsilon) \ll 1, \mu_r = 1$
 $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), $\beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)
Exact (quasi-cond) $X = \sqrt{1 + (\epsilon'' / \epsilon')^2}$
 $\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X - 1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X + 1}$
 $\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$ TM mode $H_z = 0, E_z \neq 0$
Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.
E_x form (TE) read m, n from arguments
 $E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$
coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.

TE mode <i>generator: $\tilde{H}_z$</i>	
$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TE}, \tilde{H}_y = \tilde{E}_x / Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c / f)^2}$	
TM mode <i>generator: $\tilde{E}_z$</i>	
$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TM}, \tilde{H}_y = \tilde{E}_x / Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c / f)^2}$	
TEM plane wave <i>for reference</i>	
$\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x / \eta$	
$\eta = \sqrt{\mu / \epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A}, k = \omega \sqrt{\mu \epsilon}$	
Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k \sqrt{1 - (f_c / f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c / f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c / √ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} / (2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} / (2b)
TE ₂₀	f ₂₀ = u _{p0} / a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}. In Z_{TE}/Z_{TM} and u_g, f_c = f_{mn} of the excited mode.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) result is unitless	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)	
Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10 ⁸ (m/s), m, n (unitless).	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn} / f)^2}$ (m/s), $u_p u_g = u_{p0}^2$
 $t = L / u_g$ (s) travel time through guide of length L
Higher modes have lower u_g ⇒ arrive later.

BOUNDARY

η_i, θ_t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = |Γ|²
k_i = ω√ε_{r,i}/c (rad/m)
α₂ (Np/m), β₂ (rad/m): lossy
z = 0 — boundary plane (m)
η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: **E** ⊥ this plane
||: **E** || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z_{TE}/TM — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
η = η₀ / √ε_r

VELOCITIES

u_{p0} = c / √ε_r — bulk phase vel.
u_p = u_{p0} / √(1 - (f_c / f)²)
u_g = u_{p0} √(1 - (f_c / f)²)
u_p u_g = u_{p0}²
t = L / u_g — travel time
c = 3 × 10⁸ m/s

POWER & UNITS

%P_r, %P_t — pwr refl./trans. (unitless)
%P_r = 100 |Γ|²
%P_t = 100 |τ|² η₁ / η₂
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε₀ = 8.85 × 10⁻¹² F/m
μ₀ = 4π × 10⁻⁷ H/m

DIPOLE TYPE BY LENGTH

STEP 0 — always start here $\lambda = \text{wavelength (m)}, l = \text{antenna rod length (m)}, c = 3 \times 10^8 \text{ m/s}$		
$\lambda = \frac{c}{f} \text{ (m)}$	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
= 1	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$	
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$	$\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field	
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$	
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$.	
Max @ broadside $\theta_{\text{max}} = \pi/2$; D exact $D = 1.5$ (1.76 dB)	
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$	

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l	
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$	
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001 \text{ rad} \rightarrow \text{tiny} \rightarrow 0$.	
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$	

Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$	
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$	$S_0 = \frac{15I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$	
$D = 1.64$, $R_{\text{rad}} = 36.5 \Omega$, $\theta_{\text{max}} = \pi/2$.	
Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, shortcut form	
$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$	
$D = 1.64$, $R_{\text{rad}} = 73 \Omega$, $\theta_{\text{max}} = \pi/2$.	

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$						
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$	$\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$					
°	rad	$\sin \theta$	$\cos \theta$	$\sin^2 \theta$	$\cos^2 \theta$	
0	0	0	1	0	1	
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4	
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2	
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4	
90	$\pi/2$	1	0	1	0	
180	π	0	-1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Setup N identical elements along axis, spacing d	
elem pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$	
Equal phase ($\psi_i = 0$) $\Rightarrow$ broadside beam ($\theta_{\text{max}} = \pi/2$).	
Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$.	
Pattern multiplication $S_e = \text{single element pattern}$	
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$	
Array factor (general) $k = 2\pi/\lambda$	
$F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$	
Uniform $A_i = 1$, equal phase closed form	
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$, $\gamma = kd \cos \theta$	
Peak: $F_a^{\text{max}} = N^2$ at $\gamma = 0$ (broadside, $\theta = \pi/2$). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$.	
HPBW (broadside, uniform) use Nd , not $(N-1)d$	
$\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{Nd}$	
Grating lobes appear if $d > \lambda$ (broadside). Avoid.	

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c = 3 \times 10^8 \text{ m/s}$
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377 \Omega$
$S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
a — coef. on θ^2 in Gaussian
I_0 — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ — rad. efficiency; $G = \xi D$
$D_{\text{dB}} = 10 \log_{10} D$

BEAM PARAMETERS — β , Ω_p , D

Normalize always start here		
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)		
Beamwidth β at threshold X		
Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)		
Gaussian-only shortcut ($F = e^{-a\theta^2}$). For non-Gaussian F , solve $F(\theta_X) = X$ numerically:		
Threshold	$X = 10 \text{ dB}/10$	Gauss. θ_X
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
Pattern solid angle Ω_p recognize pattern, look up		
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$ (sr)		
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$.		

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F = 1$)	$4\pi \approx 12.57$
Hertzian $\sin^2 \theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2/\ln 2$
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$
Other (use integral)	numerical

TI-36X Pro: [2nd][d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2π .

Directivity D always $D = 4\pi/\Omega_p$; common shortcuts below	
$D = \frac{4\pi}{\Omega_p}$ (unitless)	$D_{\text{dB}} = 10 \log_{10} D$
Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p/\lambda^2$. 2-axis HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$. Circular: $D \approx 4\pi/\beta^2$.	
Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$	
$G = \xi D$	$G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.	

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$	$\lambda/4$ (gnd)	1.64	2.15	36.6
monopole				
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2/8\pi \text{ W}$. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad} , R_{rad} halved. Lossless antenna ($\xi = 1$) $\Rightarrow G = D$. So in Friis, use G_t , $G_r = \text{these } D \text{ values (e.g. half-wave } G = 1.64)$.

EFFECTIVE AREA A_e

Definition antenna's RX cross section (m ²)	
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter d	
$A_p = l d = \frac{\lambda}{2} d$ (half-wave)	
Inverse: D from A_e	
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
Hertzian special case $D = 1.5$	
$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2$ (m ²)	
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.	

FRIIS TRANSMISSION FORMULA

Pwr density at RX , then RX pwr G_t, G_r linear gains; $\lambda = c/f$	
$S_r = \frac{G_t P_t}{4\pi R^2}$ (W/m ²);	$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$ (W)
Equivalent: $P_r = S_r A_{e,r}$.	
Solve for R max range	
$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$ (m)	
Equivalent form using A_e recv. area form	
$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$, $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$	
Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.	
General form (off-axis) when patterns NOT peak-aligned	
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$	
F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.	
Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.	
3. If TX is “half-wave dipole” use $G_t = 1.64$. 4. Plug into Friis.	
Use linear G , not dB. Convert FIRST.	

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity: G gain, D directivity, P_r/P_t , SNR, etc.			
$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$		$X_{\text{lin}} = 10^{X_{\text{dB}}/10}$	
e.g. $D_{\text{dB}} = 10 \log_{10} D$; $G_{\text{dB}} = 10 \log_{10} G$;		$SNR_{\text{dB}} = 10 \log_{10} (P_r/P_N)$.	
dB linear	dB linear		
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: $3 \text{ dB} \Leftrightarrow \times 2$, $10 \text{ dB} \Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver	
$P_N = k_B T_{\text{sys}} B$ (W)	
$k_B = 1.38 \times 10^{-23} \text{ J/K}$; T_{sys} system noise temp (K); B receiver bandwidth (Hz).	
Signal-to-noise ratio	
$SNR = \frac{P_{\text{rec}}}{P_N}$ (unitless)	$SNR_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs $SNR > 10 \text{ dB}$ for usable signal.	

APERTURE ANTENNAS (horn, dish)

Directivity (any shape, uniform illum.)			
$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$			
Directivity from beamwidths any aperture			
$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$			
HPBW and D by aperture shape uniform illum.; β in rad			
Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$		$\beta_x \approx 0.88\lambda/\ell_x$	$D \approx 4\pi/(\beta_x\beta_y)$
ℓ_y		$\beta_y \approx 0.88\lambda/\ell_y$	$= 4\pi\ell_x\ell_y/\lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi\ell^2}{\lambda^2}$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx \frac{4\pi}{\beta^2} = \frac{\pi^2 d^2}{\lambda^2}$
Scaling rules $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$			
Change	New D	New β	
Double ($A_p \rightarrow 2A_p$)	area $D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$	
Double freq ($\lambda \rightarrow \lambda/2$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/2$	

FRIIS / dB

P_t, P_r — TX/RX power (W)
G_t, G_r — linear gains
$S_r = G_t P_t / (4\pi R^2)$ (W/m ²)
$P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB = $\times 2$, 10 dB = $\times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m ²)
$A_e = \lambda^2 D / (4\pi)$ (m ²)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88\lambda/\ell$ (rect/sq.)
$\beta \approx 1.02\lambda/d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd \cos \theta$
$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
$F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.88\lambda/(Nd)$ (broadside)